

Quiz 2 - Results



Attempt 1 of 1

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Attempt Score 3 / 6 - 50 %

Overall Grade (Highest Attempt) 3 / 6 - 50 %

Question 1

1 / 1 point

Chebyshev's inequality states that at least $1 - \frac{1}{k^2}$ of the data lies within k standard deviations of the mean. What is the smallest whole-number value of k needed to guarantee that at least 80% of the data lies within that range?

- ☐ 1
- ☐ 2
- ☒ 3
- ☐ 4

Question 2

0 / 1 point

Which statement is correct?

- ☐ A statistically significant correlation must be strong.
- ☐ A strong correlation always means one variable causes the other.

- ☐ A small correlation can be statistically significant when the sample size is large.
- ✗ ☒ Correlation is statistically significant only when $r > 0.5$.

Question 3

0 / 1 point

1. What is the possible range of the Pearson correlation coefficient r ?

- ✗ ☒ 0 to 1
- ☐ -1 to 1
- ☐ $-\infty$ to $+\infty$
- ☐ -100 to 100

Question 4

1 / 1 point

Which statement about a probability density function (PDF) and cumulative distribution function (CDF) is correct?

- ☐ The PDF and CDF always contain different information
- ☐ The CDF gives the probability at exactly one value only
- ✓ ☒ The CDF is obtained by accumulating the PDF, and the PDF can be recovered from the CDF by differentiation
- ☐ The PDF must always increase as the random variable increases

Question 5

0 / 1 point

If the correlation between two variables is $r = 0.7$, approximately what fraction of the variance is explained?

- ✗ ☒ 25
- ☐ 36

- ☐ 49
- ☐ 64

Question 6

1 / 1 point

A model has 91% accuracy and another has 91.5% accuracy. Across different train/test splits, accuracy commonly varies by $\pm 3\%$. What is the most reasonable interpretation?

- ☐ The second model is definitely better
- ✓ ☒ The 0.5% difference may simply be caused by variance
- ☐ Both models must have exactly the same performance
- ☐ Variance does not matter when comparing models

Done